On the saturability of p-adic Lie groups

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Introduction

In this poster, we briefly discuss the notion of a p-valuation on a p-adic Lie group, and recall the Moy-Prasad filtration groups $\mathbb{G}(F)_{x,r}$ for $\mathbb{G}$ a reductive group over a finite extension $F/\mathbb{Q}_p$, where x is a point in the Bruhat-Tits building of $\mathbb{G}$. We show that, given $p > 2$ and a mild condition on a maximal torus of $\mathbb{G}$, the associated filtration on $\mathbb{G}(F)_{x,\frac{1}{p-1}+}$ is always a saturated p-valuation, which allows one to compute its Lazard Lie $\mathbb{F}_p[P]$-algebra $\operatorname{gr}\mathbb{G}(F)_{x,\frac{1}{p-1}+}$. In particular, if p is sufficiently large, this endows the pro-p Iwahori subgroup of $\mathbb{G}(F)$ with a saturated p-valuation, showing it is p-torsion-free. This generalizes work done in [4] on the split case and [7] on the semisimple case, and generalizes the Lie algebra computations done in [1].

Generalities on p-valuations

The following definitions involving p-valuations and their Lie algebras can be found in [5].

Let G be a pro-p group. A **p-valuation** on G is a function $\omega : G \setminus \{1\} \to (0,\infty)$ satisfying for any $g, h \in G$:

1. $\omega(g) > \frac{1}{p-1}$,
2. $\omega(g^{-1}h) \geq \min\{\omega(g), \omega(h)\}$,
3. $\omega([g,h]) \geq \omega(g) + \omega(h)$,
4. $\omega(g^p) = \omega(g) + 1$.

We write $G_v = \{g \in G \mid \omega(g) \geq v\}$ and $G_{v+} = \{g \in G \mid \omega(g) > v\}$.

Given a p-valued pro-p group (G, ω), one constructs an $\mathbb{F}_p[P]$-Lie algebra

$$\operatorname{gr} G = \bigoplus_{v>0} G_v/G_{v+},$$

where P acts via

$$P(gG_{v+}) = g^p G_{(v+1)+} \in G_{v+1}/G_{(v+1)+}.$$

We say that (G, ω) is **saturated** if

$$\omega(g) > \frac{p}{p-1} \qquad \Longrightarrow \qquad g = h^p \text{ for some } h \in G.$$

If (G, ω) is saturated, then there is a filtered Lie $\mathbb{Z}_p$-algebra $L_\omega(G)_{\frac{1}{p-1}+}$ endowed with mutually inverse $\log$ and $\exp$ maps

$$L_\omega(G)_{\frac{1}{p-1}+} \underset{\log}{\overset{\exp}{\rightleftarrows}} G$$

that preserve the respective filtrations.

The following result is likely standard, however the author could not find a precise reference and thus wrote a proof himself.

A Moy-Prasad-like proposition

The log and exp maps induce isomorphisms of Lie $\mathbb{F}_p[P] \cong \operatorname{gr}\mathbb{Z}_p$-algebras

$$\operatorname{gr} L_\omega(G)_{\frac{1}{p-1}+} = \bigoplus_{r>\frac{1}{p-1}} L_\omega(G)_r/L_\omega(G)_{r+} \underset{\operatorname{gr}\log}{\overset{\operatorname{gr}\exp}{\rightleftarrows}} \bigoplus_{r>\frac{1}{p-1}} G_r/G_{r+} = \operatorname{gr} G.$$

Moy-Prasad filtrations

The following setup will be used for the rest of the poster.

Let $\mathbb{G}$ be a connected reductive group over a finite extension $F/\mathbb{Q}_p$, where $p > 2$. Let $\mathbb{T} \leq \mathbb{G}$ be a maximally F-split maximal F-torus, which we assume is **weakly induced**; that is, there is a basis of $X^*(T)$ on which the wild inertia subgroup of $\operatorname{Gal}(\overline{F}/F)$ acts via a permutation action. Let $\mathbb{S} \leq \mathbb{T}$ be a maximal F-split torus, $\Phi = \Phi(\mathbb{S}, \mathbb{G})$ its relative root group. Write $G = \mathbb{G}(F)$, $T = \mathbb{T}(F)$, $U_\alpha = \mathbb{U}_\alpha(F)$ for $\alpha \in \Phi^{\text{nd}}$ the root groups, $Z = Z_{\mathbb{G}}(\mathbb{S})(F)$. Let $\mathcal{B}(\mathbb{G})$ denote the (semisimple) Bruhat-Tits building of $\mathbb{G}$.

For $x \in \mathcal{B}(G)$ and $r \geq 0$, let $G_{x,r}$ denote the rth term of the Moy-Prasad filtration, which is the closed subgroup of G generated by Z_r and $U_{\alpha,x,r}$ for all $\alpha \in \Phi^{\text{nd}}$ (see [2, Definition 7.3.3] for details). Write also

$$G_{x,r+} = \bigcup_{s>r} G_{x,s}.$$

Main theorem on the saturation of $G_{x,\frac{1}{p-1}+}$ (P.)

Assume $p > 2$ and that $\mathbb{T}$ is weakly induced. For any $x \in \mathcal{B}(\mathbb{G})$, there is a saturated p-valuation on $G_{x,\frac{1}{p-1}+}$ defined by

$$\omega(g) = \min\left\{ r > \frac{1}{p-1} \,\middle|\, g \in G_{x,r} \right\}.$$

It is immediate that this is compatible with the Iwahori decomposition

$$\prod_{\alpha\in\Phi^{-,\text{nd}}} U_{\alpha,x,\frac{1}{p-1}+} \times Z_{\frac{1}{p-1}+} \times \prod_{\alpha\in\Phi^{+,\text{nd}}} U_{\alpha,x,\frac{1}{p-1}+} \xrightarrow{\sim} G_{x,\frac{1}{p-1}+}.$$

We note that this generalizes [4, Proposition 3.5], which is a special case of this result when $\mathbb{G}$ is split, x is the barycenter of the dominant base alcove, and p is sufficiently large (so that $G_{x,\frac{1}{p-1}+} = G_{x,0+}$ is the pro-p Iwahori), as observed in [1, Remark 3.5].

The Lie algebra

For $r \geq 0$, let $\mathcal{G}_{x,r}$ be the smooth $\mathcal{O}_F$-group scheme associated to $x \in \mathcal{B}(\mathbb{G})$ which satisfies $\mathcal{G}_{x,r}(\mathcal{O}_F) = G_{x,r}$, as in [2, Section 13.3]. We write

$$\mathfrak{g}_{x,r} = \operatorname{Lie}(\mathcal{G}_{x,r})(\mathcal{O}_F),$$

which is a Lie $\mathcal{O}_F$-algebra. Then we have the following result.

Theorem on the Lie $\mathbb{Z}_p$-algebra (P.)

There is a natural isomorphism of filtered Lie $\mathbb{Z}_p$-algebras

$$L_\omega(G_{x,\frac{1}{p-1}+})_{\frac{1}{p-1}+} \cong \mathfrak{g}_{x,\frac{1}{p-1}+}.$$

Notably, after making suitable identifications, these wind up being equal as Lie $\mathbb{Z}_p$-subalgebras of the Lie algebra $\operatorname{Lie}(\mathbb{G})(F)$.

Taking the associated graded map yields an isomorphism of Lie $\mathbb{F}_p[P] \cong \operatorname{gr}\mathbb{Z}_p$-algebras

$$\operatorname{gr} L_\omega(G_{x,\frac{1}{p-1}+})_{\frac{1}{p-1}+} \cong \operatorname{gr}\mathfrak{g}_{x,\frac{1}{p-1}+},$$

which combines with the previous proposition to yield the following.

Corollary about the usual Moy-Prasad isomorphism (P.)

The Moy-Prasad isomorphism

$$\mathrm{MP}_x : \bigoplus_{r>\frac{1}{p-1}} G_{x,r}/G_{x,r+} \xrightarrow{\sim} \bigoplus_{r>\frac{1}{p-1}} \mathfrak{g}_{x,r}/\mathfrak{g}_{x,r+},$$

a priori only known to be isomorphism of $\mathbb{F}_p$-vector spaces (see [2, Section 13.5] for details), fits into a commutative diagram

$$\begin{tikzcd}
\operatorname{gr} G_{x,\frac{1}{p-1}+} \arrow[rr, "\mathrm{MP}_x"] \arrow[dr, "\operatorname{gr}\log"'] & & \operatorname{gr}\mathfrak{g}_{x,\frac{1}{p-1}+} \\
& \operatorname{gr} L_\omega(G_{x,\frac{1}{p-1}+})_{\frac{1}{p-1}+} \arrow[ur, "\sim"'] &
\end{tikzcd}$$

and hence is an isomorphism of Lie $\mathbb{F}_p[P] \cong \operatorname{gr}\mathbb{Z}_p$-algebras.

The above is a conceptual generalization of much of [1, Section 3], which only deals with the split case (as in [4]), and proves the analogous result using a computation-heavy argument that would be hard to generalize.

Example applications

As always, assume $p > 2$ and $\mathbb{T}$ is weakly induced.

As a corollary, we can conclude the following for any $x \in \mathcal{B}(\mathbb{G})$, which should be compared to [3, Corollary C.4] in the appendix by Koziol and Schwein:

Group-theoretic consequences

1. $G_{x,\frac{1}{p-1}+}$ is a p-torsion-free pro-p group.
2. For any $r \geq 1$, $G_{x,r}$ is a uniform pro-p group.
3. For any $r > \frac{1}{p-1}$, $G_{x,r+1} = \{g^p \mid g \in G_{x,r}\}$.

We also get the following as an immediate consequence of [6, Theorem 5.5], where k is any field of characteristic p, viewed as a trivial representation of $G_{x,\frac{1}{p-1}+}$:

Spectral sequence

There is a spectral sequence

$$E_1^{r,s} = H^{r,s}\left(\operatorname{gr}\left(\mathfrak{g}_{x,\frac{1}{p-1}+}/\mathfrak{g}_{x,\frac{p}{p-1}+}\right), k\right) \implies H^{r+s}\left(G_{x,\frac{1}{p-1}+}, k\right),$$

where the E_1-page has terms given by graded Lie $\mathbb{F}_p$-algebra cohomology, converging to the continuous group cohomology of $G_{x,\frac{1}{p-1}+}$.

References

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